

PROJECT REPORT



MULTI-LINE INSURANCE POLICY AND CLAIMS MANAGEMENT SYSTEM

SALESFORCE IMPLEMENTATION

Sample Academic Project Documentation

Project Role	Salesforce Developer
Platform	Salesforce CRM
Technology	Lightning Experience, Flow, Apex, SOQL
Domain	General Insurance – Multi-Line Policies & Claims
Project Status	Sample / Academic Implementation

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1. INTRODUCTION

The Multi-Line Insurance Policy and Claims Management System is a Salesforce-based solution designed to centralize insurance customer, policy, coverage, premium, claim, payment, document, and service information in one platform. The solution supports multiple insurance lines such as Motor, Health, Life, Property, and Travel insurance.

The system uses Salesforce custom objects, relationships, validation rules, record-triggered Flows, approval processes, Apex classes and triggers, reports, dashboards, and role-based security. These capabilities are intended to reduce repetitive manual work and provide a connected view of policy and claim operations.

Business Overview

- Maintain customer and contact information in a centralized Salesforce environment.
- Create and manage policies across multiple insurance product lines.
- Track policy status, coverage, premium, renewal dates, and related documents.

2. PURPOSE AND OBJECTIVES

Purpose

The purpose of the system is to provide a centralized Salesforce application for managing multi-line insurance policies and claims while improving data consistency, workflow visibility, and controlled access.

Objectives

- Centralize Customers, Policies, Coverages, Claims, Payments, Documents, and Agents.
- Support multiple insurance lines in a common data model.
- Automate policy renewal reminders and claim status notifications.
- Route selected claim decisions through an approval process.

3. PROBLEM STATEMENT AND IDEATION

Problem Statement

Persona	Need	Problem	Impact
Policy Administrator	Manage policies	Policy data may be maintained across different systems	Data duplication and slower service
Claims Officer	Process claims	Claim details, documents and policy information may be disconnected	Longer claim handling time
Underwriter	Review risk and coverage	Relevant policy and customer data may not be available together	Manual verification

Selected Approach – Salesforce Platform

Salesforce is used as the implementation platform because the project requires connected records, workflow automation, configurable security, Apex-based processing, and operational reporting in a single environment.

- Connected customer, policy and claim records.
- Declarative automation using Flow and Approval Processes.
- Programmatic processing using Apex when complex logic is required.
- Declarative automation using Flow and Approval Processes.

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4. REQUIREMENT ANALYSIS

Business Requirements

- Maintain customer information including name, contact details, address, and customer type.
- Maintain policy information including policy number, insurance line, start date, end date, premium, sum insured, and status.
- Maintain coverage details associated with each policy.
- Create claims against eligible policies.
- Track claim amount, claim type, claim status, incident date, assessment, and settlement information.
- Maintain claim documents and supporting evidence.
- Automate renewal reminders before policy expiry.
- Route high-value or selected claims for approval.
- Update related records based on policy and claim status.
- Provide dashboards for active policies, upcoming renewals, open claims, and settled claims.

Functional Requirements

FR-01	Create and manage customer records.
FR-02	Create and manage multi-line insurance policies.
FR-03	Associate policies with customers and agents.
FR-04	Maintain policy coverage information.
FR-05	Create and manage claims linked to valid policies.
FR-06	Track claim status and settlement details.
FR-07	Validate policy and claim fields before saving.
FR-08	Send policy renewal reminders before expiry.
FR-09	Route selected claims through approval.
FR-10	Provide reports and dashboards for management.

Non-Functional Requirements

- Usability—the system should support this requirement through appropriate Salesforce configuration and automation.
- Security—the system should support this requirement through appropriate Salesforce configuration and automation.
- Data Accuracy—the system should support this requirement through appropriate Salesforce configuration and automation.
- Reliability—the system should support this requirement through appropriate Salesforce configuration and automation.
- Maintainability—the system should support this requirement through appropriate Salesforce configuration and automation.
- Scalability—the system should support this requirement through appropriate Salesforce configuration and automation.
- Performance—the system should support this requirement through appropriate Salesforce configuration and automation.

5. SALESFORCE DATA MODEL

The proposed data model organizes insurance information into related Salesforce objects. Policy acts as a central transactional object, while Customer, Agent, Coverage, Claim, Payment, and Document records provide supporting information.

Object	Purpose
Customer	Stores policyholder/customer details.
Agent	Stores insurance agent/broker information.
Policy	Stores policy number, line, dates, premium, status and insured amount.
Coverage	Stores individual coverage details for a policy.
Claim	Stores claim registration, assessment and settlement information.
Claim Document	Stores references to supporting claim documents.
Payment	Stores premium or claim settlement payment details.

Key Relationships

- Policy → Customer: Lookup relationship.
- Policy → Agent: Lookup relationship.
- Coverage → Policy: Master-Detail relationship.
- Claim → Policy: Lookup relationship.
- Claim Document → Claim: Master-Detail relationship.
- Payment → Policy or Claim: Lookup relationship depending on payment type.

6. CUSTOM OBJECTS AND RELATIONSHIPS

Object	Example Fields
Customer	Customer Name, Email, Phone, Address, Customer Type
Agent	Agent Name, Email, Phone, Region, Status
Policy	Policy Number, Insurance Line, Start Date, End Date, Premium, Sum Insured, Status
Coverage	Coverage Name, Coverage Type, Limit, Deductible, Policy
Claim	Claim Number, Claim Type, Incident Date, Claim Amount, Status, Policy
Claim Document	Document Name, Document Type, Claim, Verification Status
Payment	Payment Type, Amount, Date, Status, Policy/Claim

Insurance Lines

The sample implementation supports a common Policy object with an Insurance Line field. Example values are Motor, Health, Life, Property, Travel, and Other. Line-specific requirements can be represented through Coverage records and configurable fields.

Claim Lifecycle

A sample lifecycle is: Draft → Submitted → Under Review → Approved/Rejected → Settled/Closed. The exact status transitions can be controlled using Flow, Approval Process, validation rules, and Apex where appropriate.

7. VALIDATION RULES AND FORMULA FIELDS

Validation Rules

- Policy End Date must be later than Policy Start Date.
- Premium must be greater than or equal to zero.
- Claim Incident Date should fall within the policy coverage period for an active policy.
- Claim Amount should not be negative.
- Mandatory claim information must be present before submission.
- Customer email should follow a valid email format.

Formula Fields

- Policy Duration – calculates the number of days between policy start and end dates.
- Days to Expiry – calculates remaining days before policy expiry.
- Claim Age – calculates elapsed days since claim submission.
- Net Claim Amount – can calculate the amount after applicable deductible fields.

8. FLOWS AND AUTOMATION

Automation	Trigger	Action
Policy Renewal Reminder	Scheduled before policy expiry	Send reminder to customer/owner
Claim Submission Flow	Claim status changes to Submitted	Validate required data and notify claims team
Claim Status Notification	Claim status updated	Notify relevant user/customer process
Policy Status Update	Policy dates/status change	Update status based on business rules
Document Verification	Document status updated	Update related claim verification indicator

Scheduled Renewal Reminder

Ascheduledorrecord-triggeredFlowcanusethePolicyEndDateasthetimesourceandsendarembinderbefore expiry. The notification can include policy number, insurance line, expiry date, and contact information.

Claim Approval Flow

Claimsmeetingconfiguredbusinessconditions canberoutedforapproval. Afterapproval, theClaimStatuscanbe updated and downstream notifications or payment processing steps can be initiated.

9. APEX CLASSES AND TRIGGERS

Apexisusedinthesampldesignwherecomplexprocessingorcross-recordvalidationisbetterhandled programmatically.

Component	Purpose
ClaimValidationHelper	Reusable validation logic for claim eligibility and business rules.
PreventDuplicateClaim	Checks for duplicate claim references according to configured criteria.
PolicyStatusHelper	Updates policy status based on dates and business conditions.
ClaimTrigger	Invokes helper logic after/before Claim insert or update.
BatchCompleteExpiredPolicies	Processes large volumes of expired policies.
SchedulePolicyRenewalCheck	Runs periodic checks for upcoming renewals.

10. CLAIMS MANAGEMENT WORKFLOW

The claims process is represented as a controlled workflow connecting Policy, Customer, Claim, Claim Document, and Payment information.

Step	Activity
1	Customer or service user registers a claim.
2	System validates the policy and required claim information.
3	Claim enters Submitted / Under Review status.
4	Claims officer reviews incident details and supporting documents.
5	Eligible claims may be routed for approval.
6	Approved claim proceeds toward settlement; rejected claim is closed with appropriate status.
7	Settlement/payment information is recorded.
8	Claim is marked Settled or Closed after completion.

Business Controls

- Claims should be linked to an appropriate policy.
- Policy status and coverage dates should be checked before claim submission.
- Required documents can be tracked through Claim Document records.
- Approval requirements can be configured according to claim amount or business category.
- Claim settlement should be recorded only after the required process is completed.

11. SECURITY AND ACCESS CONTROL

The sample Salesforce implementation uses layered security so that users receive access appropriate to their responsibilities.

- Profiles – define baseline object and field permissions.
- Roles – support record visibility through the role hierarchy.
- Permission Sets – provide additional permissions without changing the base profile.
- Organization-Wide Defaults – establish baseline record access.
- Sharing Rules – extend access to selected users or groups where required.
- Field-Level Security – restrict sensitive fields such as claim amounts, payment details, or customer information.

Role	Typical Access
Insurance Admin	Configuration and broad operational access
Policy Officer	Customer and policy management
Claims Officer	Claim creation, review and processing
Underwriter	Policy and risk-related review
Customer Service	Read/limited update access for service activities
Manager	Reports, dashboards and oversight

12. LIGHTNING APP, REPORTS AND DASHBOARDS

Lightning App – Insurance Operations

ALightningAppcanprovidenavigationforCustomers,Agents,Policies,Coverages,Claims,ClaimDocuments, Payments, Reports, and Dashboards.

Suggested Reports

- Active Policies by Insurance Line
- Policies Expiring in the Next 30 Days
- Open Claims by Status
- Claims by Insurance Line
- Claims Awaiting Approval
- Settled Claims by Month
- Premium Collection Summary

Suggested Dashboard

Component	Purpose
Policy Count by Line	Shows distribution across insurance products.
Upcoming Renewals	Highlights policies approaching expiry.
Open Claims	Shows current claims workload.
Claims by Status	Provides lifecycle visibility.
Claim Amount Summary	Summarizes claim amounts for management review.

13. DATA MIGRATION AND TESTING

Data Migration

ValidatedsampledatacanbeimportedintoSalesforceusingtheDataImportWizardorDataLoader,depending on data volume and object requirements. Data should be checked for duplicate customers, invalid policy dates, missing relationships, and inconsistent claim references before import.

Testing

Test Area	Expected Result
Customer Email Validation	Invalid email values are rejected.
Policy Date Validation	Invalid start/end date combinations are rejected.
Claim Eligibility	Claims cannot be submitted against an invalid/ineligible policy.
Renewal Reminder	Configured reminders are generated before expiry.
Claim Approval	Configured claims follow the approval process.
Security	Users can access only permitted records and fields.
Reports	Report results reflect the underlying records.

14. DEPLOYMENT AND MAINTENANCE

Deployment Strategy

For an academic implementation, the system can be developed and tested in a Salesforce Developer Edition or sandbox environment. For real-world use, configuration and code should be tested in the target environment before production deployment.

- Deploy custom objects, fields and relationships.
- Deploy validation rules, formulas and Flows.
- Deploy Approval Processes and Apex classes/triggers.
- Deploy profiles, permission sets and sharing configuration.
- Deploy Lightning App, reports and dashboards.
- Perform regression and user-acceptance testing before production release.

Maintenance Approach

- Monitor Flow failures and scheduled automation.
- Review Apex classes, triggers, batch jobs and scheduled jobs.
- Update validation and approval rules when business requirements change.
- Review user access and sharing configuration periodically.
- Maintain reports and dashboards as operational needs evolve.
- Monitor data quality and correct inconsistent records.

15. ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

Advantages

- Centralized insurance data management.
- Supports multiple insurance lines in a common platform.
- Automation reduces repetitive manual activities.
- Connected policy and claim records improve traceability.
- Reports and dashboards improve operational visibility.
- Role-based security supports controlled access.
- Salesforce architecture allows future expansion.

Limitations

- The sample is an academic design and does not represent a production insurance implementation.
- Real insurance workflows may require additional regulatory, compliance, document, payment, and integration requirements.
- Complex integrations may require external systems and additional development.
- Automation and Apex logic require ongoing monitoring and maintenance.

Future Scope

- Integration with payment gateways and external policy systems.
- Customer self-service portal and mobile access.

16. CONCLUSION

The Multi-Line Insurance Policy and Claims Management System demonstrates how Salesforce can be configured as a centralized platform for insurance operations. The sample solution connects customer, policy, coverage, claim, document, and payment information while using Salesforce automation, Apex, security, and reporting capabilities.

The implementation provides a structured foundation for managing policy lifecycles and claims workflows. The design can be extended with additional insurance products, integrations, advanced analytics, customer self-service capabilities, and production-grade compliance controls.

APPENDIX – SAMPLE COMPONENT LIST

- Custom Objects: Customer, Agent, Policy, Coverage, Claim, Claim Document, Payment
- Automation: Renewal Reminder Flow, Claim Notification Flow, Policy Status Flow
- Validation Rules: Policy Date Validation, Claim Amount Validation, Email Validation
- Approval Process: Selected Claim Approval
- Apex: ClaimValidationHelper, PreventDuplicateClaim, PolicyStatusHelper
- Batch/Scheduled Apex: BatchCompleteExpiredPolicies, SchedulePolicyRenewalCheck
- Security: Profiles, Roles, Permission Sets, OWD, Sharing Rules
- UI & Analytics: Lightning App, Reports, Dashboard

THANK YOU